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## POSITION-BASED INITIATION OF AUTO UNLOADING OPERATION

### Abstract


A control system on a material transfer vehicle includes a path planning system that plans a path from the material transfer vehicle to a container, and a navigation system that navigates the material transfer vehicle along the navigation path. An auto unloading system automatically controls the material transfer vehicle as it approaches the container, positions itself closely proximate the container, and unloads material into the container. A handoff control system uses position-based criteria to determine when to switch from controlling the material transfer vehicle using the path planning system to controlling the material transfer vehicle using the auto unload system. The handoff control system generates a control signal triggering the auto unload system to begin controlling the material transfer vehicle when the position-based criteria are met.



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## Background/Summary

FIELD OF THE DESCRIPTION

[0001] The present description relates to agricultural equipment. More specifically, the present description relates to automatically controlling a grain cart during an unloading process.

BACKGROUND

[0002] There are a wide variety of different types of agricultural equipment. Some such agricultural equipment includes agricultural harvesters. Agricultural harvesters often engage crop, process that crop to obtain harvested material, and unload the harvested material into a material transfer vehicle such as a tractor-pulled grain cart.

[0003] Once the material transfer vehicle is filled to a desired fill level a propulsion vehicle (such as the tractor or other vehicle) pulls the grain cart to a container, such as a semi-trailer or other haulage vehicle. The propulsion vehicle approaches the semi-trailer, pulls along side the semi-trailer, and then engages an unloading auger on the grain cart to unload the harvested material from the grain cart into the semi-trailer.

[0004] The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

[0005] A control system on a material transfer vehicle includes a path planning system that plans a path from the material transfer vehicle to a container, and a navigation system that navigates the material transfer vehicle along the navigation path. An auto unloading system automatically controls the material transfer vehicle as it approaches the container, positions itself closely proximate the container, and unloads material into the container. A handoff control system uses position-based criteria to determine when to switch from controlling the material transfer vehicle using the path planning system to controlling the material transfer vehicle using the auto unload system. The handoff control system generates a control signal triggering the auto unload system to begin controlling the material transfer vehicle when the position-based criteria are met.

[0006] Example 1 is a method of controlling a material transfer vehicle, the method comprising:

[0007] controlling the material transfer vehicle with a navigation system based on a navigation path generated by a path planning system; [0008] computing a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to a container; [0009] generating a handoff signal to handoff control of the material transfer vehicle to an auto unloading system based on a value of the position-based parameter; and [0010] controlling an unloading operation of the material transfer vehicle with the auto unloading system to automatically transfer material from the material transfer vehicle to a container based on the handoff signal.

[0011] Example 2 is the method of any or all previous examples wherein generating a handoff signal comprises: [0012] comparing the value of the position-based parameter to a threshold value to generate a comparison result; [0013] determining whether a handoff criterion is met based on the comparison result; and [0014] if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.

[0015] Example 3 is the method of any or all previous examples wherein computing the position-based parameter comprises: [0016] detecting a path-based distance indicative of a distance remaining along the navigation path between the material transfer vehicle and the container.

[0017] Example 4 is the method of any or all previous examples wherein generating the handoff signal comprises: [0018] comparing the path-based distance to a path-based distance threshold to generate the comparison result; [0019] determining that the handoff criterion is met when the comparison result indicates that the path-based distance meets the path-based distance threshold; and [0020] generating the handoff signal to trigger the auto unloading system to begin controlling

the material transfer vehicle.
[0021] Example 5 is the method of any or all previous examples wherein computing the position-based parameter comprises: [0022] detecting a straight line distance indicative of a distance between the material transfer vehicle and the container along a straight line.
[0023] Example 6 is the method of any or all previous examples wherein generating the handoff signal comprises: [0024] comparing the straight line distance to a straight line distance threshold to generate the comparison result; [0025] determining that the handoff criterion is met when the comparison result indicates that the straight line distance meets the straight line distance threshold; and [0026] generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0027] Example 7 is the method of any or all previous examples wherein computing the position-based parameter comprises: [0028] detecting a travel time value indicative of a remaining travel time between the material transfer vehicle and the container along the navigation path.
[0029] Example 8 is the method of any or all previous examples wherein generating the handoff signal comprises: [0030] comparing the travel time value to a travel time threshold to generate the comparison result; [0031] determining that the handoff criterion is met when the comparison result indicates that the travel time value meets the travel time threshold; and [0032] generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0033] Example 9 is the method of any or all previous examples wherein computing the position-based parameter comprises: [0034] detecting a position of the material transfer vehicle.
[0035] Example 10 is the method of any or all previous examples wherein generating the handoff signal comprises: [0036] comparing the position of the material transfer vehicle to a predefined location to generate the comparison result; [0037] determining that the handoff criterion is met when the comparison result indicates that the position of the material transfer vehicle is within a threshold distance of the predefined; and [0038] generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0039] Example 11 is the method of any or all previous examples wherein generating the handoff signal comprises: [0040] comparing the position of the material transfer vehicle to a boundary of a predefined unloading zone to generate the comparison result; [0041] determine that the handoff criterion is met when the position of the material transfer vehicle is within the boundary of the predefined unloading zone based on the comparison result; [0042] generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0043] Example 12 is an agricultural system, comprising: [0044] a navigation system on a material transfer vehicle configured to receive a navigation path and control the material transfer vehicle based on a navigation path; [0045] an auto unloading system configured to control an unloading operation of the material transfer vehicle to automatically transfer material from the material transfer vehicle to a container; [0046] a position parameter computation processor configured to compute a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to the container; and [0047] a handoff signal generator configured to generate a handoff signal to trigger the navigation system and the auto unloading system to control the material transfer vehicle based on the position-based parameter.
[0048] Example 13 is the agricultural system of any or all previous examples and further comprising: [0049] a comparison system configured to compare a value of the position-based parameter to a threshold value to generate a comparison result, wherein the handoff signal generator is configured to determine whether a handoff criterion is met based on the comparison result, and if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0050] Example 14 is the agricultural system of any or all previous examples wherein the position parameter computation processor comprises: [0051] a path distance calculator configured to detect

a path-based distance indicative of a distance remaining along the navigation path between the material transfer vehicle and the container and wherein the comparison system is configured to compare the path-based distance to a path-based distance threshold to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the path-based distance meets the path-based distance threshold.
[0052] Example 15 is the agricultural system of any or all previous examples wherein the position parameter computation processor comprises: [0053] a straight line distance calculator configured to detect a straight line distance indicative of a distance between the material transfer vehicle and the container along a straight line, wherein the comparison system is configured to compare the straight line distance to a straight line distance threshold to generate the comparison result and wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the straight line distance meets the straight line distance threshold.
[0054] Example 16 is the agricultural system of any or all previous examples wherein the position parameter computation processor comprises: [0055] a travel time calculator configured to detect a travel time value indicative of a remaining travel time between the material transfer vehicle and the container along the navigation path, wherein the comparison system is configured to compare the travel time value to a travel time threshold to generate the comparison result, and wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the travel time value meets the travel time threshold.
[0056] Example 17 is the agricultural system of any or all previous examples wherein the position parameter computation processor comprises: [0057] a location identifier configured to detect a position of the material transfer vehicle and a predefined location wherein the comparison system is configured to compare the position of the material transfer vehicle to a predefined location to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the position of the material transfer vehicle is within a threshold distance of the predefined.
[0058] Example 18 is the agricultural system of any or all previous examples wherein the position parameter computation processor comprises: [0059] a zone identifier configured to detect a position of the material transfer vehicle and a boundary of a predefined loading zone, wherein the comparison system is configured to compare the position of the material transfer vehicle to the boundary of the predefined unloading zone to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the position of the material transfer vehicle is within the boundary of the predefined unloading zone.
[0060] Example 19 is a material transfer vehicle, comprising: [0061] a propulsion subsystem; [0062] a steering subsystem; [0063] an unloading conveyor; [0064] a path planning system configured to generate a travel path from the material transfer vehicle to an unloading location; [0065] a navigation system configured to receive the travel path and control the propulsion subsystem and the steering subsystem based on a navigation path; [0066] an auto unloading system configured to control the propulsion subsystem, the steering subsystem, and the unloading conveyor to automatically transfer material from the material transfer vehicle to a container; [0067] a position parameter computation processor configured to compute a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to the container; and [0068] a handoff signal generator configured to generate a handoff signal to trigger one of the navigation system and the auto unloading system to control the material transfer vehicle based on the position-based parameter.
[0069] Example 20 is the material transfer vehicle of any or all previous examples and further comprising: [0070] a comparison system configured to compare a value of the position-based parameter to a threshold value to generate a comparison result, wherein the handoff signal

generator is configured to determine whether a handoff criterion is met based on the comparison result, and if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
[0071] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0072] FIG. **1** is a partial pictorial, partial block diagram of an agricultural system.
[0073] FIG. **2** is a partial pictorial, partial block diagram of an agricultural system using a path-based distance as a position-based handoff criterion.
[0074] FIG. **3** is a partial pictorial, partial block diagram of an agricultural system using straight-line distance as a position-based handoff criterion.
[0075] FIG. **4** is a partial pictorial, partial block diagram of an agricultural system using travel time as a position-based handoff criterion.
[0076] FIG. **5** is a partial pictorial, partial block diagram of an agricultural system using a predefined unloading zone as a position-based handoff criterion.
[0077] FIG. **6** is a partial pictorial, partial block diagram of an agricultural system using a predefined location as a position-based handoff criterion.
[0078] FIG. **7** is a block diagram showing one example of a material transfer vehicle in more detail.
[0079] FIG. **8** is a flow diagram illustrating one example of the operation of a handoff control system in processing a position-based criterion to determine whether a navigation planning system or an auto unloading system should be controlling the material transfer vehicle.
[0080] FIG. **9** is a block diagram showing one example of an agricultural system deployed in a remote server environment.
[0081] FIG. **10** is a block diagram of one example of a computing environment that can be used in the architectures and systems described in other figures.
DETAILED DESCRIPTION
[0082] As discussed above, it is common for a material transfer vehicle, such as a tractor-pulled grain cart, to transfer harvested material from a harvester to a container, such as a haulage vehicle (e.g., a semi-trailer). Some current control systems on material transfer vehicles include a path planning system that plans a path from a location of the material transfer vehicle to an estimated location of the container. A navigation system automatically navigates the material transfer vehicle along the navigation path. An auto unloading system can control the material transfer vehicle to automatically unload the material from the grain cart into the container. By automatically, it is meant that, in one example, the operation or function can be performed without further human involvement except, perhaps, to initiate or authorize the operation or function.
[0083] In order to control the material transfer vehicle to automatically unload material into the container, the auto unloading system may use on-board sensors to detect the precise location and pose of the container. Using that location and pose, the auto unloading system can control the propulsion and steering subsystems on the material transfer vehicle to move the material transfer vehicle along side the container so that the material can be unloaded, such as by using an auger and spout. The auto unloading system controls the material transfer vehicle to move along the container so that a relatively even fill (or another desired fill pattern) can be achieved when transferring material from the material transfer vehicle into the container. However, it can be difficult to known

when to switch between controlling the material transfer vehicle using the path planning system and controlling the vehicle using the auto unloading system.

[0084] The present description thus describes a system in which a handoff control system detects and processes position-based criteria in order to determine when to hand control off from the path planning system to the auto unloading system. The handoff control system can use the distance between the material transfer vehicle and the container, measured along the navigation path, as the position-based criterion. The handoff control system can also use a straight-line distance between the material transfer vehicle and the container, regardless of the navigation path as the position-based criterion. The handoff control system can use an estimated travel time along the navigation path as the position-based criterion. The control system can use a predefined unloading zone as the position-based criterion. The handoff control system can also use a predefined point or a predefined location as the position-based criterion.

[0085] FIG. **1** is a partial pictorial, partial block diagram of an agricultural system **100** in which a harvester **102** is moving through a field in the direction indicated by arrow **104** and harvesting crop. The crop can be unloaded from harvester **102** into material transfer vehicle **106**. In the example shown in FIG. **1**, material transfer vehicle **106** includes a towing vehicle **108** (such as a tractor or other towing vehicle) and a grain cart **110**. Once grain cart **110** is filled to a desired level with harvested material from harvester **102**, material transfer vehicle **106** travels along a navigation path indicated by arrow **112** to position itself along side a container **118** which may include a semi-trailer **120**, or another container.

[0086] FIG. **1** also shows that agricultural system **100** can include a control system **122** which may be disposed on towing vehicle **108**, on grain cart **110**, in a remote server environment, or distributed among a plurality of different locations. Control system **102** includes path planning system **124**, auto unloading system **126**, handoff control system **128**, navigation system **130**, and other items **132**. Until position-based criteria are met, path planning system **124** calculates and updates the path **112** between material transfer vehicle **106** and container **118**. Navigation system **130** automatically navigates material transfer vehicle **106** to travel along the navigation path **112**. During that navigation, handoff control system **128** determines whether position-based handoff criteria (also referred to as position-based criterion or handoff criteria) are met and, if so, generates a handoff control signal to control auto unloading system **126** to begin controlling material transfer vehicle **106**, instead of path planning system **124** and navigation system **130**. It will be appreciated that some portions of navigation system **130** may continue to control material transfer vehicle **106** under the control of auto unloading system **126**, once a handoff takes place or auto unloading system **126** can control material transfer vehicle **126** without using navigation system **130**.

[0087] Once material transfer vehicle **106** is under the control of auto unloading system **126**, auto unloading system **126** controls material transfer vehicle **106** to move along side trailer **120**. Grain cart **110** may include a gate **134** at its lower end that can be opened and closed to allow material from grain cart **110** to move into a hopper or other collection structure where the material can be transferred by an auger **136** and spout **138** into semi-trailer **120**. Therefore, once auto unloading system **126** has controlled material transfer vehicle **106** to come into a position next to semi-trailer **120**, auto unloading system **126** can control material transfer vehicle **106** to engage a power takeoff to drive auger **136**, and to open gate **134** to allow material to be transferred from grain cart **110** into semi-trailer **120** using auger **136** and spout **138**. Auto unloading system **126** continues to move material transfer vehicle **106** along trailer **120**, to obtain a desired fill level in semi-trailer **120**. Auto unloading system **126** can also control the movement of material transfer vehicle **106** to unload material according to a predefined fill pattern (such as a front-to-back fill pattern, a back-to-front fill pattern, a multi-pass fill pattern, etc.).

[0088] In order to generate the handoff control signal, handoff control system **128** first computes a positioned-based criterion and then determines whether that criterion indicates that a handoff from path planning system **124** and navigation system **130** to auto unloading system **126** should take place. FIG. **2** shows one example in which handoff control system **128** uses, as the position-based criterion, the distance remaining between material transfer vehicle **106** and container **118** along navigation path **112**. In FIG. **2**, handoff control system **128** computes the distance along travel path **112** between material transfer vehicle **106** and container **118**. Once that distance reaches a threshold level (which may be represented in FIG. **2** as the distance between point **140** and container **118** on travel path **112**, then handoff control system **128** determines that the handoff criterion are met and generates a handoff control signal that hands off control of material transfer vehicle **106** from path planning system **124** and navigation system **130**, to auto unloading system **126**. Again, it will be appreciated that whenever auto unloading system **126** is controlling material transfer vehicle **106**, portions of navigation system **130** may be used as well. However, those portions of navigation system **130** will be under the control of auto unloading system **126**.

[0089] In another example, illustrated in FIG. **3**, handoff control system **128** may use, as the position-based criterion, the straight-line distance between material transfer vehicle **106** and container **118**. FIG. **3** shows that, from a predetermined location on container **118**, a circle of radius r is drawn around container **118**. Once material transfer vehicle **106** comes within the distance r of container **118**, then handoff control system **128** determines that the position-based criterion are met and that control of material transfer vehicle **106** should be handed off to auto unloading system **126**. This may be done, for example, when the sensors on-board material transfer vehicle **106** have a high enough accuracy to adequately control material transfer vehicle **106** once those sensors are close enough to (e.g., within distance r of) container **118**.

[0090] FIG. **4** illustrates an architecture in which handoff control system **128** uses travel time as the position-based criterion in order to determine whether control should be handed off to auto unloading system **126**. By way of example, handoff control system **128** may receive a sensor signal indicative of a current travel speed of material transfer vehicle **106** as well as a signal indicative of the remaining distance long path **112** between material transfer vehicle **106** and container **118**. Using those parameters, and perhaps an expected speed of material transfer vehicle **106** along the remainder of path **112**, handoff control system **128** computes an estimated travel time remaining for material transfer vehicle **106** to travel to an unloading position proximate container **118**. When that travel time reaches a threshold value (as illustrated by the X **142** in FIG. **4**) then handoff control system **128** determines that the position-based criterion is met and that control should be handed off to auto unloading system **126**.

[0091] FIG. **5** shows an example in which handoff control system **128** uses a pre-defined (or dynamically generated) loading zone **144** to determine whether the position-based criterion is met. For instance, loading zone **144** may be pre-marked on a map that is accessed by handoff control system **128**. In another example, loading zone **144** may be automatically or manually generated on a map once the location of container **118** is identified. Regardless of how loading zone **144** is generated, handoff control system **128** can determine that, when material transfer vehicle **106** is within the boundaries of loading zone **144**, then the position-based criterion is met and control of material transfer vehicle **106** should be handed off to auto unloading system **126**.

[0092] FIG. **6** shows an example in which handoff control system **128** uses a predefined location **146** as the position-based criterion. The predefined location **146** may be a point in a coordinate system, or a larger area within a coordinate system. The location of the predefined location **146** can be manually generated, or automatically generated. The position of predefined location **146** may be based on a known or estimated position of container **118**, or it may be generated in other ways. Handoff control system **128** determines when material transfer vehicle **106** reaches (or is within a threshold distance of) the predefined location **146**. When that happens, then handoff control system **128** determines that the position-based criterion is met and that control should be handed off to auto unloading system **126**.

[0093] FIG. **7** is a block diagram of one example of material transfer vehicle **106**, in more detail. In the example shown in FIG. **7**, towing vehicle **108** includes one or more processors or servers **148**, data store **150**, communication system **152**, sensors **154**, operator interface system **156**, path planning system **124**, handoff control system **128**, auto unloading system **126**, navigation system **130**, controllable subsystems **158**, and other towing vehicle functionality **160**. Data store **150** can include distance-related criteria **162** and other items **164**. Sensors **154** can include location sensor **166**, one or more container sensors **168**, one or more unload operation sensors **170**, and other sensors **172**. Handoff control system **128** can include data store interaction system **174**, position parameter computation processor **176**, comparison system **178**, handoff signal generator **180**, and other items **182**. Position parameter computation processor **176** can include path distance calculator **184**, straight line distance calculator **186**, travel time calculator **188**, location identifier **190**, zone identifier **192**, and other items **194**. Auto unloading system **126** can include vehicle position controller **196**, fill level/fill strategy controller **198**, unloading functionality controller (e.g., auger/spout position controller, gate position controller, PTO engagement controller, engine speed controller, etc.) **200**, and other items **202**. Controllable subsystems **158** can include propulsion subsystem **204**, steering subsystem **206**, power take off (PTO) **208**, and other controllable subsystems **210**.

[0094] In the example shown in FIG. **7**, grain cart **110** can include one or more sensors **212**, auger position actuator **214**, spout position actuator **216**, auger **136**, spout **138**, gate **134**, and other grain cart functionality **218**.

[0095] FIG. **7** also shows that towing vehicle **108** can be connected to grain cart **110** by a link **220**. Link **220** can be a mechanical link, a link that transfers mechanical power, electrical power, pneumatic power, hydraulic power, or other power. Link **220** can also be a communication link that is used for communication between towing vehicle **108** and grain cart **110**. Link **220** can include a wired link, a wireless link or other link as well.

[0096] Before describing the operation of handoff control system **128** in more detail, a description of some of the items on material transfer vehicle **106** and their operation will first be provided. The position-based criterion **162** can identify the type of position-based criterion that handoff control system **128** will use (such as any of those described above with respect to FIGS. **1-6** or other criteria). The position-based criterion **162** can also include the values for those criteria (such as the path-based distance, the straight-line distance r, the travel time, the location of the unloading zone **144**, the specific location **146**, or the value of another position-based criterion).

[0097] Communication system **152** can be used to facilitate communication of the items of towing vehicle **108** and grain cart **100** with one another. Therefore, communication system **152** can be a controller area network (CAN) bus and bus controller. Communication system **152** can also facilitate communication between towing vehicle **108** and other machines, other systems, etc. Therefore, communication system **152** can be a cellular communication system, a wide area network communication system, a local area network communication system, a near field communication system, a Bluetooth or wifi communication system, among any other communication systems or combinations of communication systems.

[0098] Sensors **154** sense variables and provide sensor signals indicative of the sensed variables. Location sensor **166** senses the location of sensor **166** in a global or local coordinate system. Therefore, location sensor **166** can be a global navigation satellite system (GNSS) receiver, a dead reckoning system, a cellular triangulation system, or another location sensor. Container sensors **168** can be sensors that sense the location and pose of container **118**. The container sensors **168** may have a sensor range that is defined by the particular sensor. Container sensors **168** can include optical sensors (such as a stereo camera with an image processing system), a RADAR sensor, a LIDAR sensor, an ultra wideband sensor, an ultrasound sensor, among other sensors. Unload operation sensors **170** sense variables that are used by auto unloading system **126** to automatically control material transfer vehicle **106** to perform an auto unloading operation. Therefore, unload operation sensors **170** may include optical sensors or other sensors that sense the fill level of material in container **118**, as material is being unloaded from grain cart **110** into container **118**.

Unload operation sensors **170** may also sense the landing point of material in container **118**, as it is being transferred through auger **136** and spout **138**. The landing point can be used to adjust spout **138** to change the trajectory of material as it is exiting auger **136** and entering container **118**. Unload sensors **170** can include weight sensors, volume sensors, or any of a wide variety of other sensors that may be used by auto unloading system **126**.

[0099] Operator interface system **156** may include operator interface mechanisms, such as a steering wheel, joy sticks, pedals, levers, linkages, display screens, touch sensitive screens, actuatable elements on display screens (such as icons, links, buttons, etc.), a speaker, a microphone (such as where speech recognition and/or speech synthesis are provided) among any of a wide variety of other audio, visual, and/or haptic mechanisms that can be used to generate outputs for an operator and receive inputs from an operator. The operator may be a human operator or an automated or semi-automated system.

[0100] Path planning system **124** can receive an input from location sensor **166** indicative of a location of vehicle **108**, as well as an input indicative of a location of container **118**. The location of container **118** may be estimated, or communicated over communication system **152** from container **118**, itself. For instance, if container **118** has a GNSS receiver, the location identified by the GNSS receiver on container **118** can be transmitted to path planning system **124** using communication system **152**. Path planning system **124** generates and updates a navigation path **112** between the current location of towing vehicle **108** and container **118**. Path planning system **124** can implement any of a wide variety of path planning algorithms, such as grid-based search algorithms, sampling-based search algorithms, trajectory optimization algorithms, among others.

[0101] Navigation path **112** is provided to navigation system **130** which generates control signals to control the propulsion subsystem **204** and steering subsystem **206** of towing vehicle **208** in order to navigate towing vehicle **108** along the navigation path **112**. Navigation system **130** receives inputs, such as from location sensor **166** and other perception inputs and generates control signals to control propulsion system **204** and steering subsystem **206** to navigate towing vehicle **108** along the navigation path **112**.

[0102] In auto unloading system **126**, vehicle position controller **196** controls the position of towing vehicle **108** relative to container **118**, based upon inputs from container sensors **168** and possibly other inputs. Vehicle position controller **196** thus controls towing vehicle **108** to bring grain cart **110** along side container **118**, and sufficiently close to container **118** that material can be transferred from grain cart **110** into container **118** by auger **136** and spout **138**. Unloading functionality controller **200** generates control signals to control various items of unloading functionality during an automated unload operation. For instance, unloading functionality controller **200** can control auger position actuator **214** and spout position actuator **216** to move auger **136** and spout **138** from a transport position to a deployed, unload position. Unload functionality controller **200** can also control PTO **208** to engage the PTO, and unloading functionality controller **200** can control the engine speed of towing vehicle **108** to thus change the rate at which material is transferred from grain cart **110** to container **118**. Further, unloading functionality controller **200** can generate control signals to control gate position actuator **217** to control the position of gate **134** between its open position and closed position.

[0103] Fill level/fill strategy controller **198** can also receive an input from a fill level sensor (e.g., one of unload operation sensors **170** which may be an optical sensor mounted on auger **136** or elsewhere with a field of view inside container **118**). Fill level/fill strategy controller **198** calculates an estimated fill level at a current material landing point in container **118**. When that fill level reaches a desired fill level, then vehicle position controller **196** can nudge towing vehicle **108** forward relative to container **118** to change the landing point of the material inside container **118** to a point that has yet to be filled (or that has an insufficient fill level). Also, in one example, unloading functionality controller **200** can control the position of auger **136** and spout **138** to change the landing point of material in container **118** as well. Fill level/fill strategy controller **198** can also generate control signals to move towing vehicle **108** to execute a desired fill strategy (such as a front-to-back fill strategy in container **118**, a back-to-front fill strategy, a multi-pass fill strategy, etc.).

[0104] Propulsion subsystem **204** can include an engine and transmission that is used to drive ground engaging elements (such as wheels or tracks) on towing vehicle **108** to propel towing vehicle **108**. Propulsion system **204** can also include individual motors that drive individual ground engaging elements or sets of ground engaging elements, or other propulsion mechanisms. Steering subsystem **206** can include steerable wheels or skid steer wheels or tracks that can be used to control the heading of towing vehicle **108**. Power take off PTO **208** is illustratively a power takeoff mechanism driven by the engine of towing vehicle **108**. Therefore, by controlling engine speed, the speed of rotation of PTO **208** can also be controlled.

[0105] Data store interaction system **174** in handoff control system **128** interacts with data store **150** to obtain the distance-related criteria **162**. Thus, data store interaction system **174** can obtain any of the position-based criteria discussed above, or others.

[0106] Position parameter computation processor **176** performs any desired computations that are needed to determine whether the position-based criteria **162** are met. For instance, where the position-based criteria **162** include the path distance as described above with respect to FIG. **2**, then path distance calculator **184** calculates the distance along path **112** that remains between material transfer vehicle **106** and container **108**. Where the position-based criteria **162** include the straight line criterion described above with respect to FIG. **3**, then straight line distance calculator **186** calculates the straight line distance between material transfer vehicle **106** and container **118**. Where the position-based criteria **162** include a travel time criterion as discussed above with respect to FIG. **4**, then travel time calculator **188** calculates the estimated travel time remaining for material transfer vehicle **106** to travel along navigation path **112** to reach container **118**. Where the position-based criteria **162** include an unloading zone **144** as discussed above with respect to FIG. **5**, then zone identifier **192** identifies the location of the boundaries of the unloading zone **144** that is to be used as the distance-based criterion. Where the distance-based criteria **162** include a specific location as discussed above with respect to FIG. **6**, then location identifier **190** identifies that specific location that is to be used as the position-based criterion.

[0107] Once the position-based parameter is computed by position parameter computation processor **176**, then comparison system **178** compares the parameter value(s) to threshold value(s) to determine whether a position-based criterion is met. For instance, where the path distance calculator **184** calculates the remaining path distance along path **112**, and that is to be used as the position-based criterion, then comparison system **178** compares the path distance calculated by calculator **184** to a threshold path distance to determine when the material transfer vehicle **106** is within the path distance threshold (e.g., **140** as described in FIG. **2** above). If so, then the path distance criterion is met and handoff signal generator **180** generates a handoff signal indicating that auto unloading system **126** should begin controlling towing vehicle **108**. Path distance calculator **184** can continuously or intermittently calculate new path distance values and comparison system **178** can compare those new path distance values against the threshold value on a continuous or intermittent basis as well.

[0108] Where the position-based criterion is the straight line distance r described above with respect to FIG. **3**, then straight line distance calculator **186** calculates the straight line distance between material transfer vehicle **106** and container **118** and comparison system **178** compares that value to a threshold value r to determine whether material transfer vehicle **106** is within the threshold straight line distance of container **118**. If so, then the position-based criterion is met and handoff signal generator **180** generates a handoff signal again indicating that auto unloading system **126** is to begin controlling towing vehicle **108**. The straight line distance can be continuously or intermittently updated and compared by calculator **186** and comparison system **178**, respectively.

[0109] Where the position-based criterion is a remaining travel time criterion, then travel time calculator **188** calculates the estimated travel time remaining for material transfer vehicle **106** to travel the remaining distance along travel path **112** to reach container **118**. Again, the travel time can be estimated based on estimated speeds of towing vehicle **108**, based upon the current speed of towing vehicle **108**, among other things. The travel time is compared by comparison system **178** against a travel time threshold value to determine whether the remaining travel time calculated by calculator **188** meets the travel time threshold. If so, handoff signal generator **180** generates a handoff signal again indicating that unloading system **126** is to begin controlling towing vehicle **108**.

[0110] Where the position-based criterion is a specific location, then location identifier **190** identifies that location and provides that location to comparison system **178** which compares that location against a current location of towing vehicle **108** output by location sensor **166**. If the two locations are within a threshold distance of one another, then this indicates that the position-based criterion is met, and handoff signal generator **180** again generates a handoff signal indicting that auto unloading system **126** is to begin controlling towing vehicle **108**. Where the position-based criterion includes an unloading zone (as described by zone **144** with respect to FIG. **5**), then zone identifier **192** identifies the location of the boundaries of zone **144** and comparison system **178** determines whether a current location of towing vehicle **108** output by location sensor **166** is within the unloading zone **144**. If so, then the position-based criterion is met and handoff signal generator **180** generates a handoff signal indicating that auto unloading system **126** is to begin controlling towing vehicle **108**.

[0111] As mentioned, all of the values calculated by the position parameter computation processor **176** can be intermittently or continuously updated, and comparison system **178** can make intermittent or continuous comparisons as well.

[0112] On grain cart **110**, sensors **212** can include any of a wide variety of different sensors. Sensors **212** can sense the position of auger **136**, spout **138**, and/or gate **134**. Sensors **212** can generate an output indicative of a fill level of material in grain cart **110** and/or container **118**.

[0113] Auger position actuator **214** drives the position of auger **136** relative to a frame of grain cart **110**, such as between a stowed or transport position and a deployed, unload position. Spout position actuator **216** controls the position of spout **138** relative to auger **136** to change the trajectory of material exiting auger **136**. Gate position actuator **217** moves gate **134** between its open position and closed position.

[0114] FIG. **8** is a flow diagram illustrating one example of the operation of handoff control system **128**. Data store interaction system **174** first obtains the position-based parameters or criteria that are to be met in order to handoff control to auto unloading system **126**. Obtaining the position parameters or criteria is indicated by block **240** in the flow diagram of FIG. **8**. As discussed above, the position-based criteria can include a path-based distance **242**, a straight line distance **244**, a travel time **246**, a predefined location **248**, or a predefined unloading zone **250**, or other distance-based criteria **252**.

[0115] During operation, it is first assumed that path planning system **124** has generated a navigation path **112** from material transfer vehicle **106** to the location of container **118**, as indicated by block **254**. The path **112** can be generated from material transfer vehicle **106** to the location of the container **118**, itself, as indicated by block **256** or to a predefined point **258** that may be generated based upon an estimate of where container **118** will be. The path **112** can be to a predefined or dynamically generated unloading zone **144**, or to another location generated in another way, as indicated by block **260**.

[0116] Navigation system **130** then begins controlling controllable subsystems **158** to navigate towing vehicle **108** along the navigation path **112**. Using navigation system **130** to navigate towing vehicle **108** along the navigation path **112** is indicated by block **262** in the flow diagram of FIG. **8**.

[0117] Position parameter computation processor **176** then computes any parameters that are needed in order to determine whether the distance-based criteria are met. Performing position processing, in order to handoff control of towing vehicle **108** to auto unloading system **126** is indicated by block **264** in the flow diagram of FIG. **8**. The position-based parameters are first computed, as indicated by block **266**. Thus, path distance calculator **184** can calculate the path distance. Straight line distance calculator **186** can calculate the straight line distance. Travel time calculator **188** can calculate the travel time. Location identifier **190** can identify the predetermined location, and/or zone identifier **192** can identify a predetermined unloading zone.

[0118] Comparison system **178** then compares the calculated parameter to a threshold value in order to determine whether the position-based criterion is met. Thus, comparison system **178** compares the path distance to a threshold distance, as indicated by block **268**. Comparison system **178** can compare the straight line distance to a threshold distance, as indicated by block **270**. Comparison system **178** can compare the travel time to a threshold travel time as indicated by block **272**. Comparison system **178** can compare a current location to a predefined location, as indicated by block **274**. Comparison system **178** can compare a current location to the location of boundaries of a predefined zone, as indicated by block **276**. Comparison system **178** can compare other computed parameters against other threshold values, as indicated by block **278**. Based on the comparison value, comparison system **178** generates an output indicative of whether the handoff criterion (e.g., the position-based criterion) is met. This determination is indicated by block **280** in the flow diagram of FIG. **8**.

[0119] If the handoff criterion is not met, then the navigation system **130** continues to control towing vehicle **108** and processing reverts to block **262** in the flow diagram of FIG. **8**. However, if, at block **280**, it is determined that the position-based (e.g., handoff) criterion is met, then handoff signal generator **180** generates an output signal triggering auto unloading system **126** to begin controlling material transfer vehicle **106**, as indicated by block **282** in the flow diagram of FIG. **8**. Auto unloading system **126** then controls the material transfer vehicle **106** to automatically perform an unloading operation, as indicated by block **284**. The unloading operation can be performed, as discussed above, based on inputs from unloading operation sensors **170**, as indicated by block **286**, or in a wide variety of other ways, as indicated by block **288**.

[0120] It can thus be seen that the present description describes a system that uses position-based parameters or criteria to determine when to switch between controlling a material transfer vehicle with a path planning and navigation system, and with an auto unloading system that may use on-board sensors to perform more precise positioning and unloading operations. The position-based criteria can be generated statically, generated and modified dynamically, or generated in other ways. This improves the accuracy and efficiency with which a handoff is made between the two control systems.

[0121] It will be noted that the above discussion has described a variety of different systems, components, calculators, identifiers, sensors, generators, and/or logic. It will be appreciated that such systems, components, calculators, identifiers, sensors, generators, and/or logic can be comprised of hardware items (such as processors and associated memory, or other processing components, some of which are described below) that perform the functions associated with those systems, components, calculators, identifiers, sensors, generators, and/or logic. In addition, the systems, components, calculators, identifiers, sensors, generators, and/or logic can be comprised of software that is loaded into a memory and is subsequently executed by a processor or server, or other computing component, as described below. The systems, components, calculators, identifiers, sensors, generators, and/or logic can also be comprised of different combinations of hardware, software, firmware, etc., some examples of which are described below. These are only some examples of different structures that can be used to form the systems, components, calculators, identifiers, sensors, generators, and/or logic described above. Other structures can be used as well.

[0122] The present discussion has mentioned processors and servers. In one example, the processors and servers include computer processors with associated memory and timing circuitry, not separately shown. The processors and servers are functional parts of the systems or devices to which they belong and are activated by, and facilitate the functionality of the other components or items in those systems.

[0123] Also, a number of user interface (UI) displays have been discussed. The UI displays can take a wide variety of different forms and can have a wide variety of different user actuatable input mechanisms disposed thereon. For instance, the user actuatable input mechanisms can be text boxes, check boxes, icons, links, drop-down menus, search boxes, etc. The mechanisms can also be actuated in a wide variety of different ways. For instance, the mechanisms can be actuated using a point and click device (such as a track ball or mouse). The mechanisms can be actuated using hardware buttons, switches, a joystick or keyboard, thumb switches or thumb pads, etc. The mechanisms can also be actuated using a virtual keyboard or other virtual actuators. In addition, where the screen on which the mechanisms are displayed is a touch sensitive screen, the mechanisms can be actuated using touch gestures. Also, where the device that displays them has speech recognition components, the mechanisms can be actuated using speech commands.

[0124] A number of data stores have also been discussed. It will be noted the data stores can each be broken into multiple data stores. All can be local to the systems accessing them, all can be remote, or some can be local while others are remote. All of these configurations are contemplated herein.

[0125] Also, the figures show a number of blocks with functionality ascribed to each block. It will be noted that fewer blocks can be used so the functionality is performed by fewer components. Also, more blocks can be used with the functionality distributed among more components.

[0126] FIG. **9** is a block diagram of agricultural system **100**, shown in FIG. **1**, except that its elements are disposed in a cloud computing architecture **500**. Cloud computing provides computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various embodiments, cloud computing delivers the services over a wide area network, such as the internet, using appropriate protocols. For instance, cloud computing providers deliver applications over a wide area network and they can be accessed through a web browser or any other computing component. Software or components of system **100** as well as the corresponding data, can be stored on servers at a remote location. The computing resources in a cloud computing environment can be consolidated at a remote data center location or they can be dispersed. Cloud computing infrastructures can deliver services through shared data centers, even though they appear as a single point of access for the user. Thus, the components and functions described herein can be provided from a service provider at a remote location using a cloud computing architecture. Alternatively, the components and functions can be provided from a conventional server, or the components and functions can be installed on client devices directly, or in other ways.

[0127] The description is intended to include both public cloud computing and private cloud computing. Cloud computing (both public and private) provides substantially seamless pooling of resources, as well as a reduced need to manage and configure underlying hardware infrastructure.

[0128] A public cloud is managed by a vendor and typically supports multiple consumers using the same infrastructure. Also, a public cloud, as opposed to a private cloud, can free up the end users from managing the hardware. A private cloud may be managed by the organization itself and the infrastructure is typically not shared with other organizations. The organization still maintains the hardware to some extent, such as installations and repairs, etc.

[0129] In the example shown in FIG. **9**, some items are similar to those shown in other FIGS. and they are similarly numbered. FIG. **9** specifically shows that handoff control system **128** can be located in cloud **502** (which can be public, private, or a combination where portions are public while others are private). Therefore, vehicles **106**, **118**, other machines **504**, and/or other systems **506** can access those systems through cloud **502**. FIG. **9** also shows an operator **505** that can interact with vehicle **106**.

[0130] FIG. **9** also depicts another example of a cloud architecture. FIG. **9** shows that it is also contemplated that some elements of system **100** can be disposed in cloud **502** while others are not. By way of example, handoff control system **128** and data store **150** can be disposed outside of cloud **502**, and accessed through cloud **502**. Regardless of where the items are located, the items can be accessed directly by the other vehicles and systems, through a network (either a wide area network or a local area network), the items can be hosted at a remote site by a service, or the items can be provided as a service through a cloud or accessed by a connection service that resides in the cloud. All of these architectures are contemplated herein.

[0131] It will also be noted that system **100**, or portions of it, can be disposed on a wide variety of different devices. Some of those devices include servers, desktop computers, laptop computers, tablet computers, or other mobile devices, such as palm top computers, cell phones, smart phones, multimedia players, personal digital assistants, etc.

[0132] FIG. **10** is one example of a computing environment in which system **100**, or parts of it, (for example) can be deployed. With reference to FIG. **10**, an example system for implementing some embodiments includes a computing device in the form of a computer **810** programmed to operate as discussed above. Components of computer **810** may include, but are not limited to, a processing unit **820** (which can comprise processors or servers from previous FIGS.), a system memory **830**, and a system bus **821** that couples various system components including the system memory to the processing unit **820**. The system bus **821** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus also known as Mezzanine bus. Memory and programs described with respect to other FIGS. can be deployed in corresponding portions of FIG. **10**.

[0133] Computer **810** typically includes a variety of computer readable media. Computer readable media can be any available media that can be accessed by computer **810** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. Computer storage media includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer **810**. Communication media typically embodies computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer readable media.

[0134] The system memory **830** includes computer storage media in the form of volatile and/or nonvolatile memory such as read only memory (ROM) **831** and random access memory (RAM) **832**. A basic input/output system **833** (BIOS), containing the basic routines that help to transfer information between elements within computer **810**, such as during start-up, is typically stored in ROM **831**. RAM **832** typically contains data and/or program modules that are immediately accessible to and/or presently being operated on by processing unit **820**. By way of example, and not limitation, FIG. **10** illustrates operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

[0135] The computer **810** may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. **10** illustrates a hard disk drive **841** that reads from or writes to non-removable, nonvolatile magnetic media, and an optical disk drive **855** that reads from or writes to a removable, nonvolatile optical disk **856** such as a CD ROM or other optical media. Other removable/non-removable, volatile/nonvolatile computer storage media that can be used in the exemplary operating environment include, but are not limited to, magnetic tape cassettes, flash memory cards, digital versatile disks, digital video tape, solid state RAM, solid state ROM, and the like. The hard disk drive **841** is typically connected to the system bus **821** through a non-removable memory interface such as interface **840**, and optical disk drive **855** are typically connected to the system bus **821** by a removable memory interface, such as interface **850**.

[0136] Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (ASICs), Application-specific Standard Products (ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

[0137] The drives and their associated computer storage media discussed above and illustrated in FIG. **10**, provide storage of computer readable instructions, data structures, program modules and other data for the computer **810**. In FIG. **10**, for example, hard disk drive **841** is illustrated as storing operating system **844**, application programs **845**, other program modules **846**, and program data **847**. Note that these components can either be the same as or different from operating system **834**, application programs **835**, other program modules **836**, and program data **837**. Operating system **844**, application programs **845**, other program modules **846**, and program data **847** are given different numbers here to illustrate that, at a minimum, they are different copies.

[0138] A user may enter commands and information into the computer **810** through input devices such as a keyboard **862**, a microphone **863**, and a pointing device **861**, such as a mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit **820** through a user input interface **860** that is coupled to the system bus, but may be connected by other interface and bus structures, such as a parallel port, game port or a universal serial bus (USB). A visual display **891** or other type of display device is also connected to the system bus **821** via an interface, such as a video interface **890**. In addition to the monitor, computers may also include other peripheral output devices such as speakers **897** and printer **896**, which may be connected through an output peripheral interface **895**.

[0139] The computer **810** is operated in a networked environment using logical connections to one or more remote computers, such as a remote computer **880**. The remote computer **880** may be a personal computer, a hand-held device, a server, a router, a network PC, a peer device or other common network node, and typically includes many or all of the elements described above relative to the computer **810**. The logical connections depicted in FIG. **10** include a local area network (LAN) **871** and a wide area network (WAN) **873**, but may also include other networks. Such networking environments are commonplace in offices, enterprise-wide computer networks, intranets and the Internet.

[0140] When used in a LAN networking environment, the computer **810** is connected to the LAN **871** through a network interface or adapter **870**. When used in a WAN networking environment, the computer **810** typically includes a modem **872** or other means for establishing communications over the WAN **873**, such as the Internet. The modem **872**, which may be internal or external, may be connected to the system bus **821** via the user input interface **860**, or other appropriate mechanism. In a networked environment, program modules depicted relative to the computer **810**, or portions thereof, may be stored in the remote memory storage device. By way of example, and

not limitation, FIG. **10** illustrates remote application programs **885** as residing on remote computer **880**. It will be appreciated that the network connections shown are exemplary and other means of establishing a communications link between the computers may be used.
[0141] It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein.
[0142] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

## Claims

**1**. A method of controlling a material transfer vehicle, the method comprising: controlling the material transfer vehicle with a navigation system based on a navigation path generated by a path planning system; computing a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to a container; generating a handoff signal to handoff control of the material transfer vehicle to an auto unloading system based on a value of the position-based parameter; and controlling an unloading operation of the material transfer vehicle with the auto unloading system to automatically transfer material from the material transfer vehicle to a container based on the handoff signal.
**2**. The method of claim 1 wherein generating a handoff signal comprises: comparing the value of the position-based parameter to a threshold value to generate a comparison result; determining whether a handoff criterion is met based on the comparison result; and if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**3**. The method of claim 2 wherein computing the position-based parameter comprises: detecting a path-based distance indicative of a distance remaining along the navigation path between the material transfer vehicle and the container.
**4**. The method of claim 3 wherein generating the handoff signal comprises: comparing the path-based distance to a path-based distance threshold to generate the comparison result; determining that the handoff criterion is met when the comparison result indicates that the path-based distance meets the path-based distance threshold; and generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**5**. The method of claim 2 wherein computing the position-based parameter comprises: detecting a straight line distance indicative of a distance between the material transfer vehicle and the container along a straight line.
**6**. The method of claim 5 wherein generating the handoff signal comprises: comparing the straight line distance to a straight line distance threshold to generate the comparison result; determining that the handoff criterion is met when the comparison result indicates that the straight line distance meets the straight line distance threshold; and generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**7**. The method of claim 2 wherein computing the position-based parameter comprises: detecting a travel time value indicative of a remaining travel time between the material transfer vehicle and the container along the navigation path.
**8**. The method of claim 7 wherein generating the handoff signal comprises: comparing the travel time value to a travel time threshold to generate the comparison result; determining that the handoff criterion is met when the comparison result indicates that the travel time value meets the travel time threshold; and generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.

**9**. The method of claim 2 wherein computing the position-based parameter comprises: detecting a position of the material transfer vehicle.
**10**. The method of claim 9 wherein generating the handoff signal comprises: comparing the position of the material transfer vehicle to a predefined location to generate the comparison result; determining that the handoff criterion is met when the comparison result indicates that the position of the material transfer vehicle is within a threshold distance of the predefined; and generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**11**. The method of claim 9 wherein generating the handoff signal comprises: comparing the position of the material transfer vehicle to a boundary of a predefined unloading zone to generate the comparison result; determine that the handoff criterion is met when the position of the material transfer vehicle is within the boundary of the predefined unloading zone based on the comparison result; generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**12**. An agricultural system, comprising: a navigation system on a material transfer vehicle configured to receive a navigation path and control the material transfer vehicle based on a navigation path; an auto unloading system configured to control an unloading operation of the material transfer vehicle to automatically transfer material from the material transfer vehicle to a container; a position parameter computation processor configured to compute a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to the container; and a handoff signal generator configured to generate a handoff signal to trigger the navigation system and the auto unloading system to control the material transfer vehicle based on the position-based parameter.
**13**. The agricultural system of claim 12 and further comprising: a comparison system configured to compare a value of the position-based parameter to a threshold value to generate a comparison result, wherein the handoff signal generator is configured to determine whether a handoff criterion is met based on the comparison result, and if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.
**14**. The agricultural system of claim 13 wherein the position parameter computation processor comprises: a path distance calculator configured to detect a path-based distance indicative of a distance remaining along the navigation path between the material transfer vehicle and the container and wherein the comparison system is configured to compare the path-based distance to a path-based distance threshold to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the path-based distance meets the path-based distance threshold.
**15**. The agricultural system of claim 13 wherein the position parameter computation processor comprises: a straight line distance calculator configured to detect a straight line distance indicative of a distance between the material transfer vehicle and the container along a straight line, wherein the comparison system is configured to compare the straight line distance to a straight line distance threshold to generate the comparison result and wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the straight line distance meets the straight line distance threshold.
**16**. The agricultural system of claim 13 wherein the position parameter computation processor comprises: a travel time calculator configured to detect a travel time value indicative of a remaining travel time between the material transfer vehicle and the container along the navigation path, wherein the comparison system is configured to compare the travel time value to a travel time threshold to generate the comparison result, and wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the travel time value meets the travel time threshold.
**17**. The agricultural system of claim 13 wherein the position parameter computation processor

comprises: a location identifier configured to detect a position of the material transfer vehicle and a predefined location wherein the comparison system is configured to compare the position of the material transfer vehicle to a predefined location to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the comparison result indicates that the position of the material transfer vehicle is within a threshold distance of the predefined.

**18**. The agricultural system of claim 13 wherein the position parameter computation processor comprises: a zone identifier configured to detect a position of the material transfer vehicle and a boundary of a predefined loading zone, wherein the comparison system is configured to compare the position of the material transfer vehicle to the boundary of the predefined unloading zone to generate the comparison result, wherein the handoff signal generator is configured to determine that the handoff criterion is met when the position of the material transfer vehicle is within the boundary of the predefined unloading zone.

**19**. A material transfer vehicle, comprising: a propulsion subsystem; a steering subsystem; an unloading conveyor; a path planning system configured to generate a travel path from the material transfer vehicle to an unloading location; a navigation system configured to receive the travel path and control the propulsion subsystem and the steering subsystem based on a navigation path; an auto unloading system configured to control the propulsion subsystem, the steering subsystem, and the unloading conveyor to automatically transfer material from the material transfer vehicle to a container; a position parameter computation processor configured to compute a position-based parameter indicative of a characteristic of a position of the material transfer vehicle relative to the container; and a handoff signal generator configured to generate a handoff signal to trigger one of the navigation system and the auto unloading system to control the material transfer vehicle based on the position-based parameter.

**20**. The material transfer vehicle of claim 19 and further comprising: a comparison system configured to compare a value of the position-based parameter to a threshold value to generate a comparison result, wherein the handoff signal generator is configured to determine whether a handoff criterion is met based on the comparison result, and if the handoff criterion is met, generating the handoff signal to trigger the auto unloading system to begin controlling the material transfer vehicle.